Estimating Rupture Dimensions of Three Major Earthquakes in Sichuan, China, for Early Warning and Rapid Loss Estimates

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We playback local and regional on-scale strong motion waveforms recorded during the 2008 M_W 7.9 Wenchuan, 2013 M_W 6.6 Lushan, and 2017 M_W 6.5 Jiuzhaigou earthquakes to study the possible performance of the Finite Fault Rupture Detector (FinDer) algorithm for earthquake early warning and rapid loss estimates after devastating earthquakes for the current layout of the China Strong Motion Network. As an important supplement to the text, this file contains plots of the vertical records of the strong ground motion of the three earthquakes studied and the FinDer line-source models at different time steps from event origin. Source parameters used for the loss estimates of the Wenchuan earthquake and the PAGER results are also shown here.

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Table S1. Source model parameters for the Wenchuan earthquake used for loss estimates								
Model	**Lon**	**Lat**	**Endpoint1-Lon**	**Endpoint1-Lat**	**Endpoint2-Lon**	**Endpoint2-Lat**	***M***	**Depth/km**
Point-source	103.42	31.01	--	--	--	--	7.9	16
Line-source from inversion	104.5	32.02	103.05	30.78	105.94	33.27	7.9	16
Line-source from FinDer	104.51	32.08	103.33	31.24	105.69	32.92	8	16

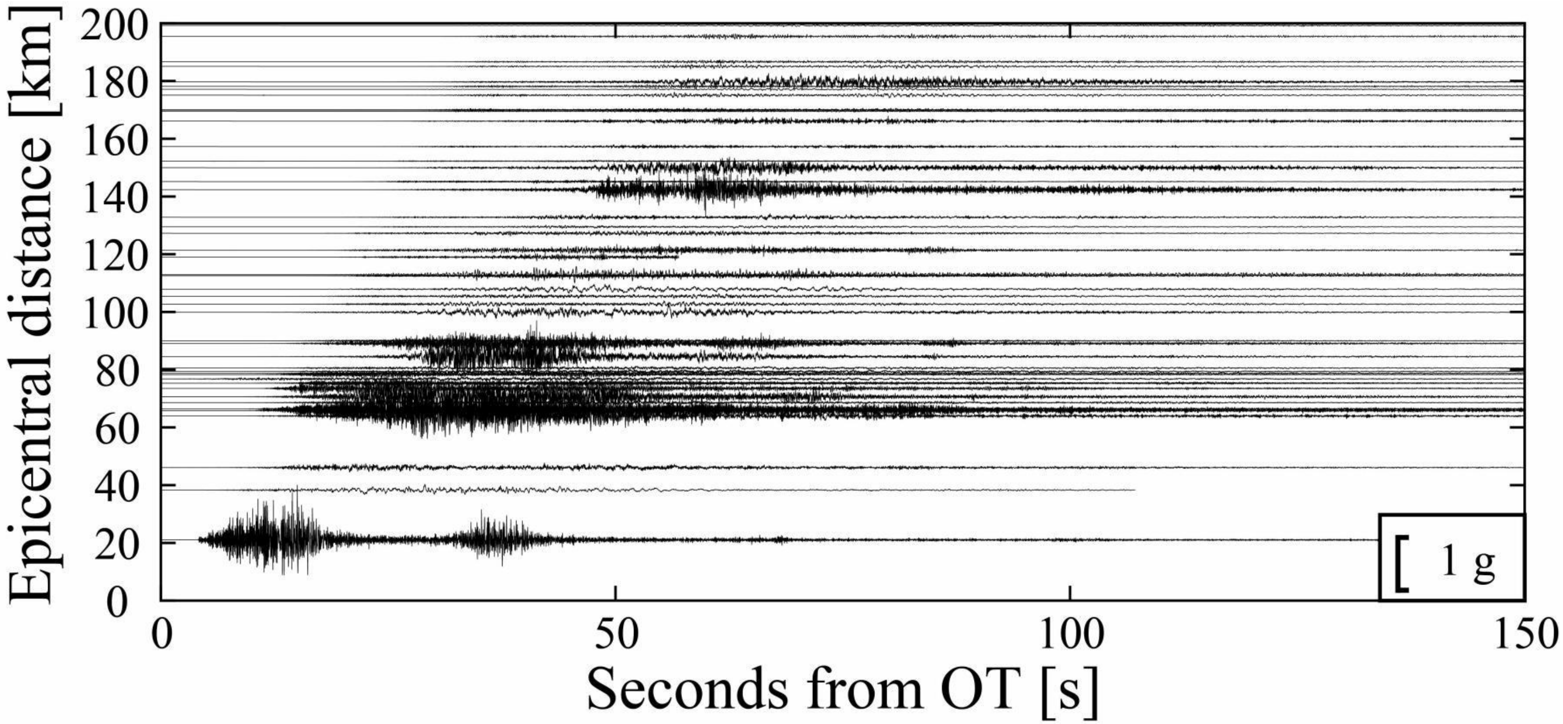

Figure S1. Corrected waveforms (vertical component) of the 2008 M_W 7.9 Wenchuan earthquake

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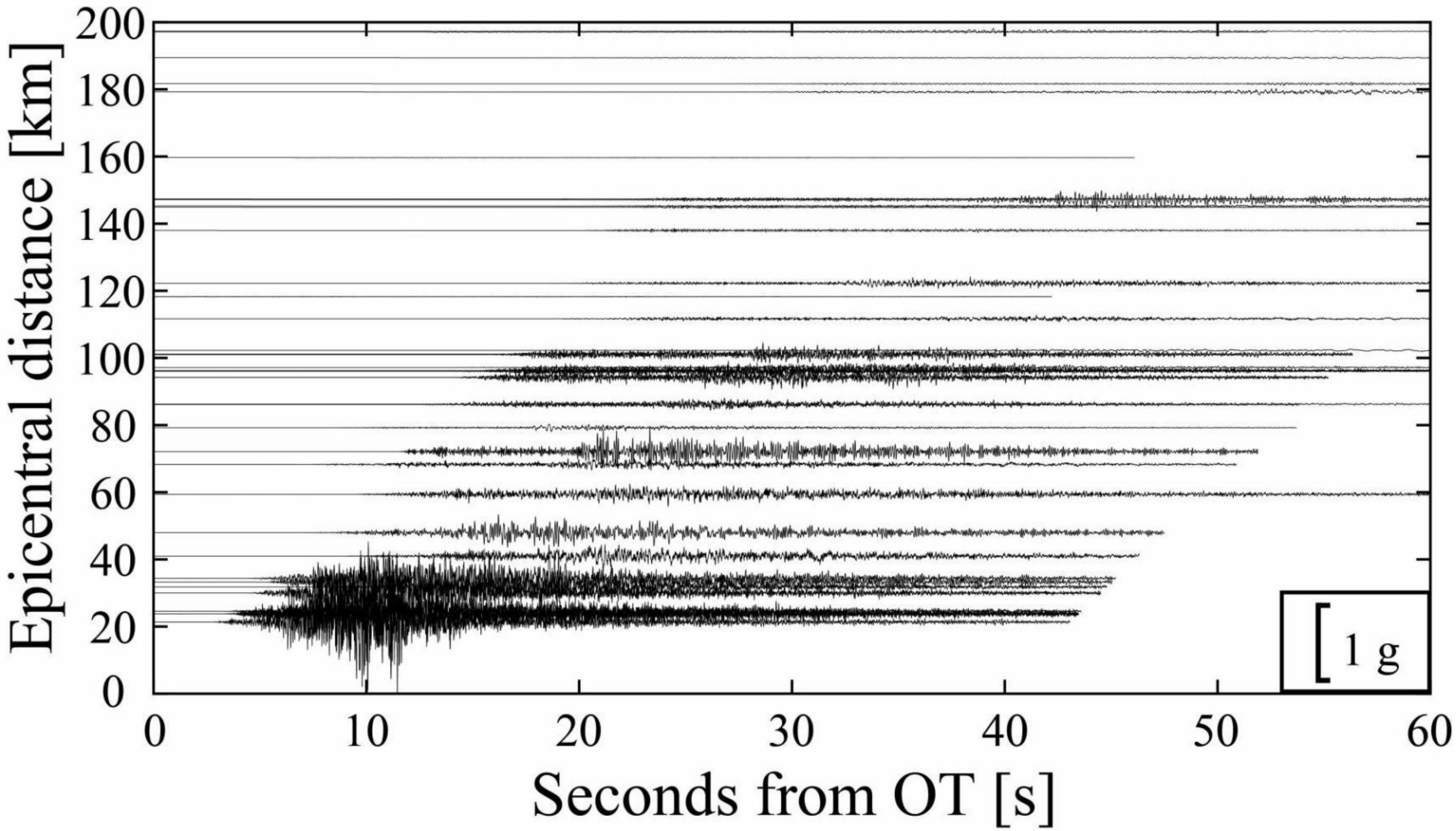

Figure S2. Corrected waveforms (vertical component) of the 2013 M_W 6.6 Lushan earthquake

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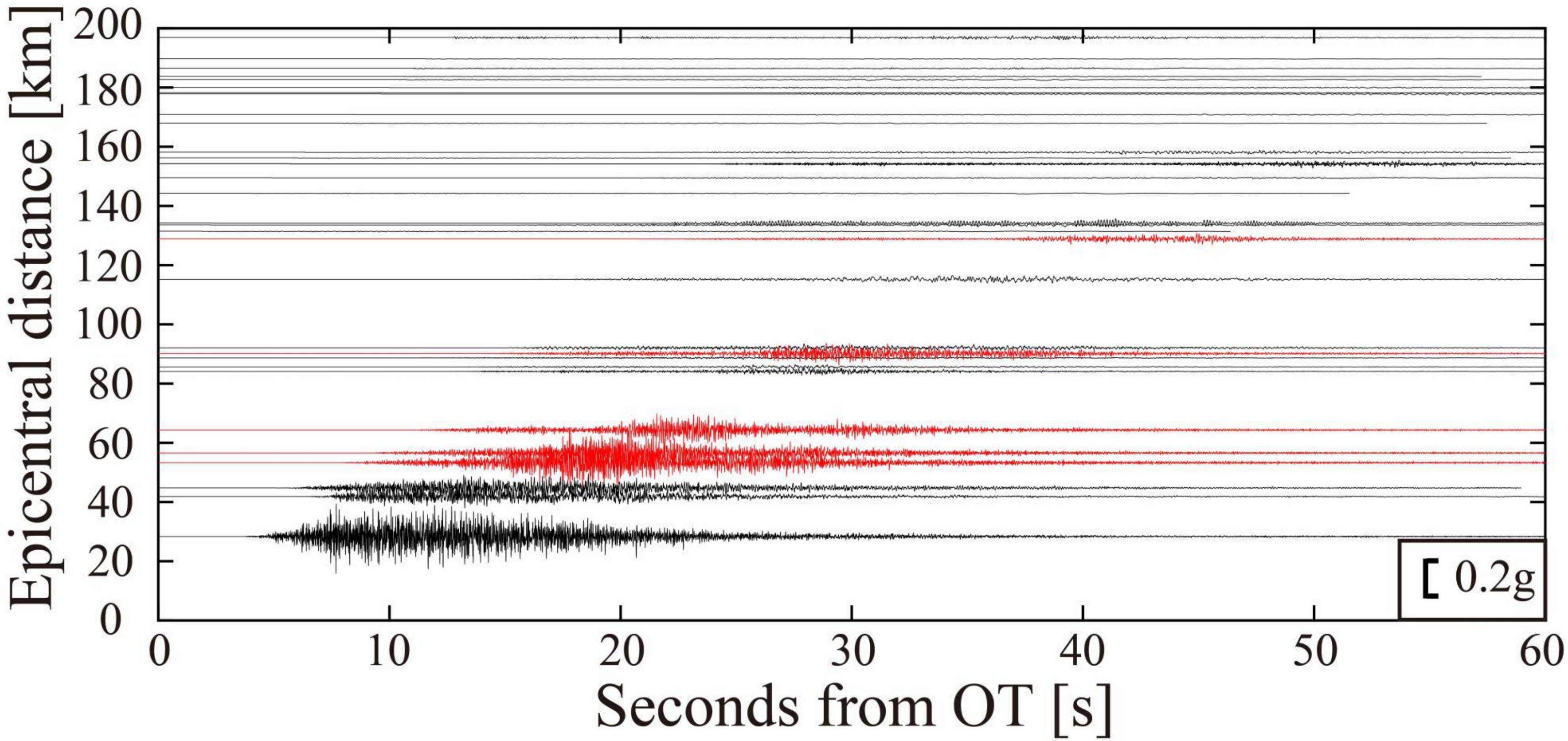


Figure S3. Corrected waveforms (vertical component) of the 2017 M_W 6.5 Jiuzhaigou earthquake. Red colored traces are simulated (see main text).

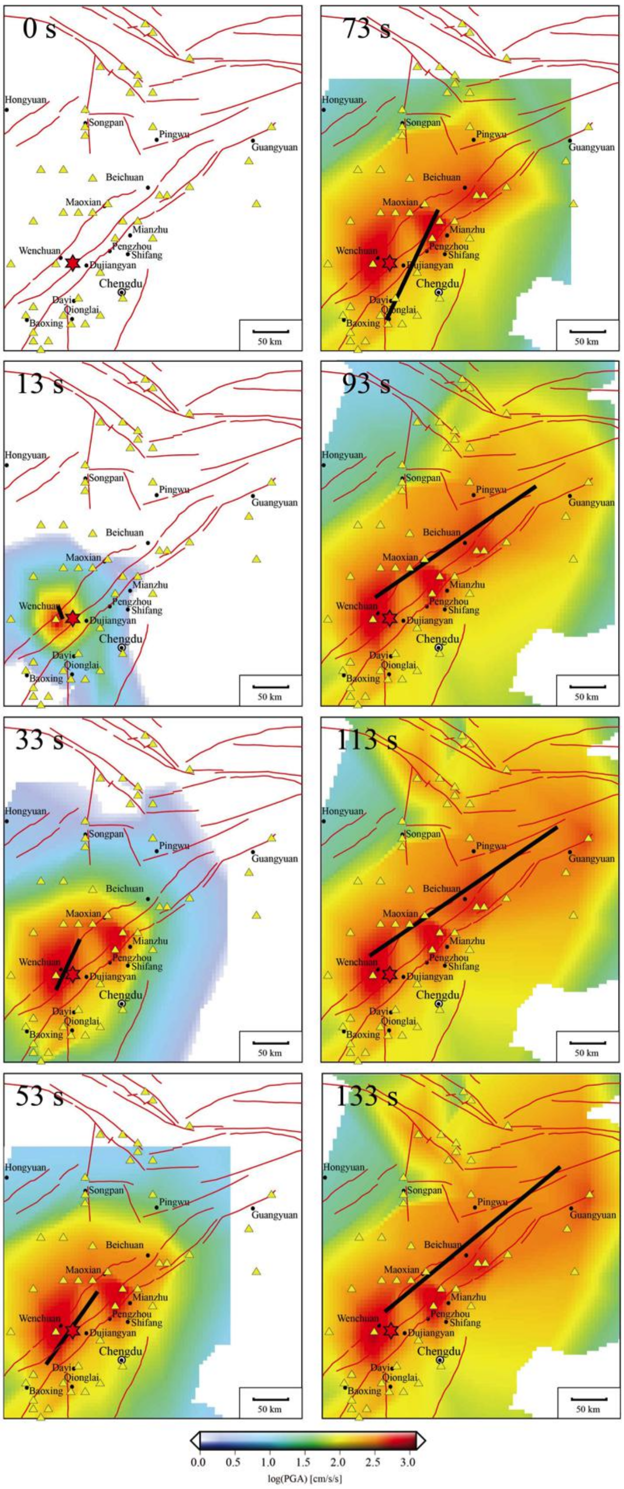


Figure S4. FinDer line-source (black line) for the M_W 7.9 Wenchuan earthquake at different time steps from event origin. Colors show interpolated observed peak ground acceleration (PGA) at each time step. The red star marks the epicenter (CENC). Yellow triangles are stations used in FinDer (follows Figure 1).

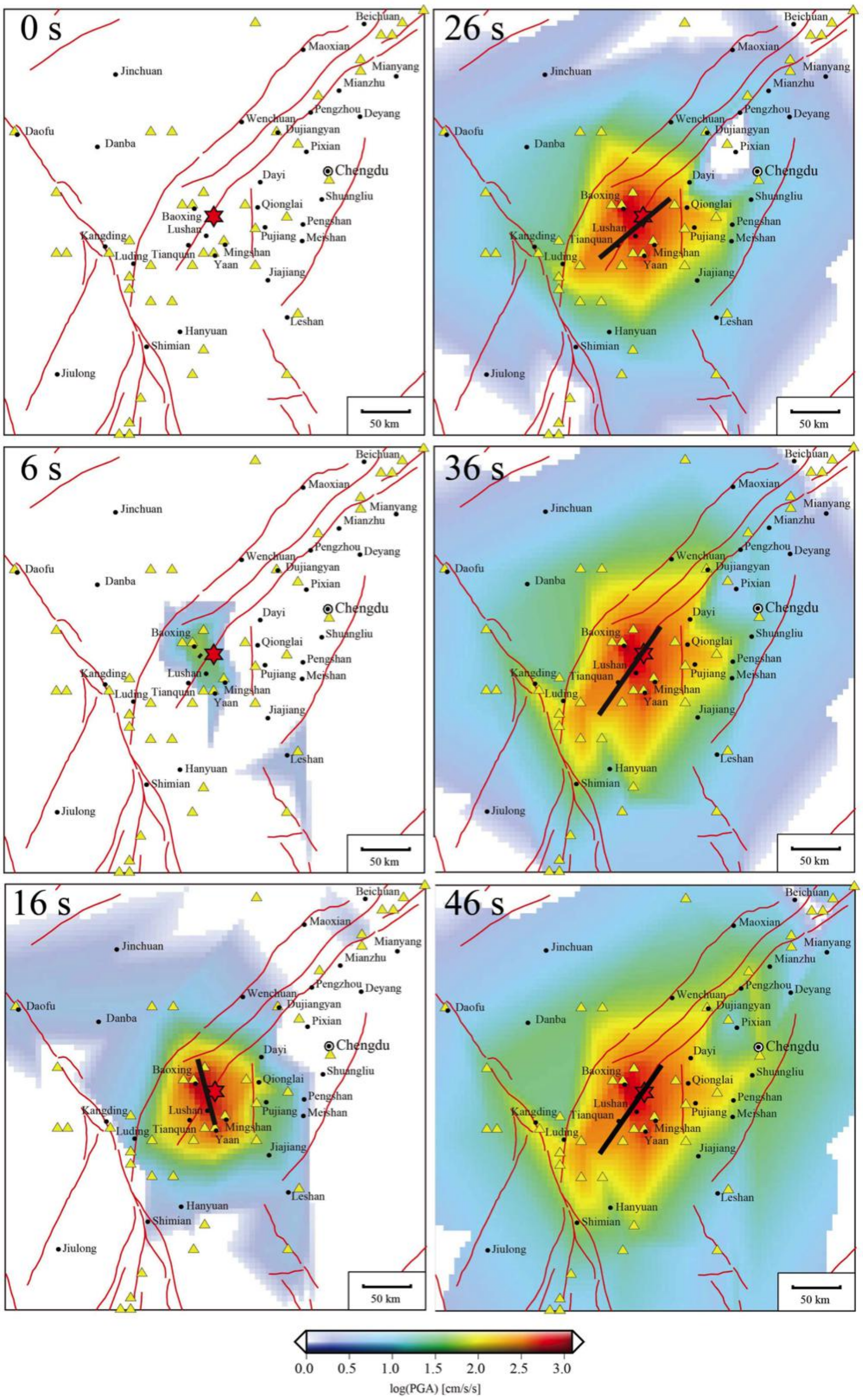


Figure S5. FinDer line-source (black line) for the M_W 6.6 Lushan earthquake at different time steps from event origin. Follows Figure S4.

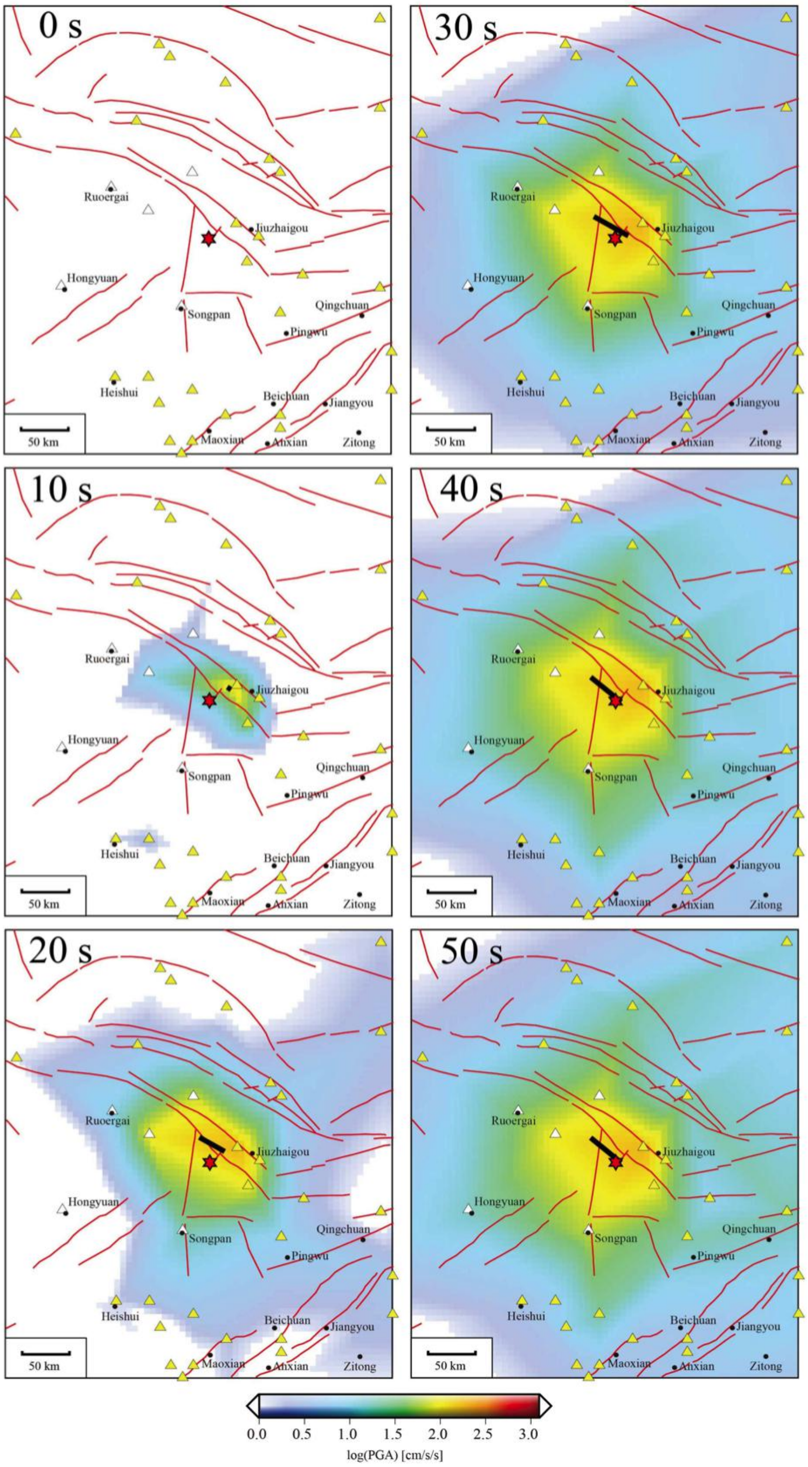


Figure S6. FinDer line-source (black line) for the M_W6.5 Jiuzhaigou earthquake at different time steps from event origin. Follows Figures S4 and S5.

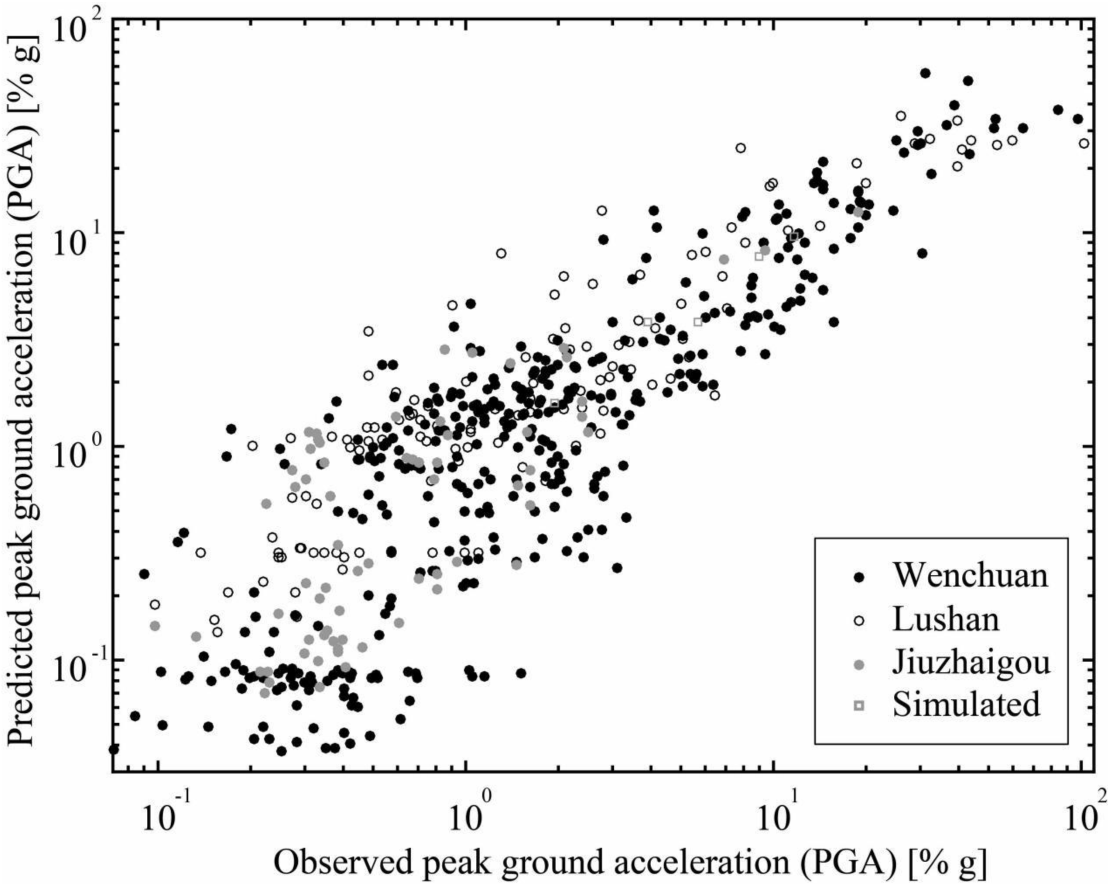
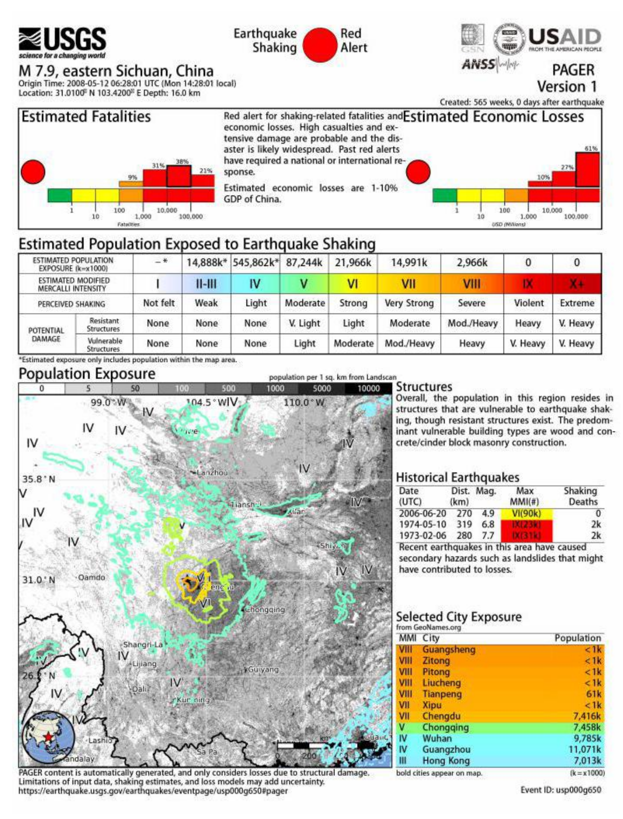
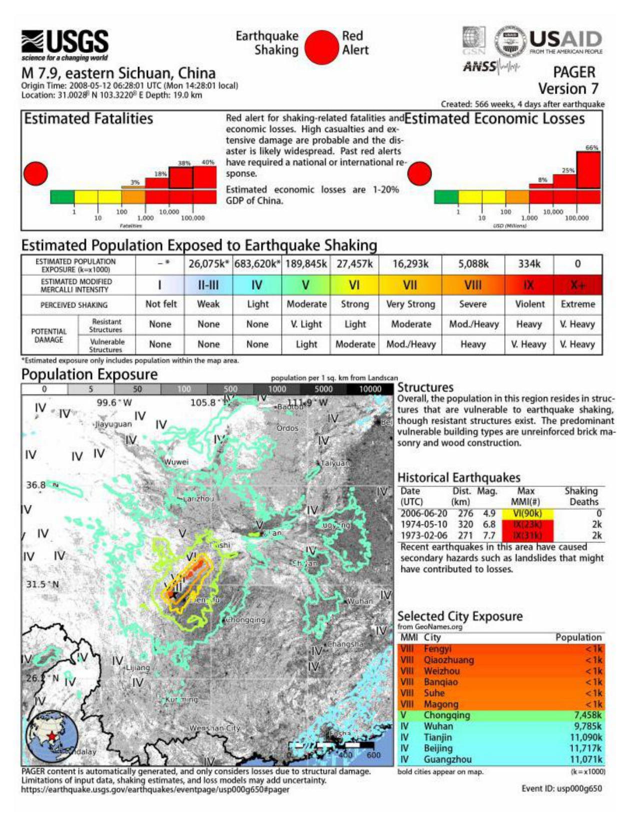


Figure S7. Observed *vs* predicted PGA values estimated from GMPEs after Cua and Heaton (2009) using the distance to the final FinDer line-source for the M_W 7.9 Wenchuan, 2013 M_W 6.6 Lushan, and 2017 M_W 6.5 Jiuzhaigou earthquakes.

Point-source	Line-source from inversion	Line-source from FinDer





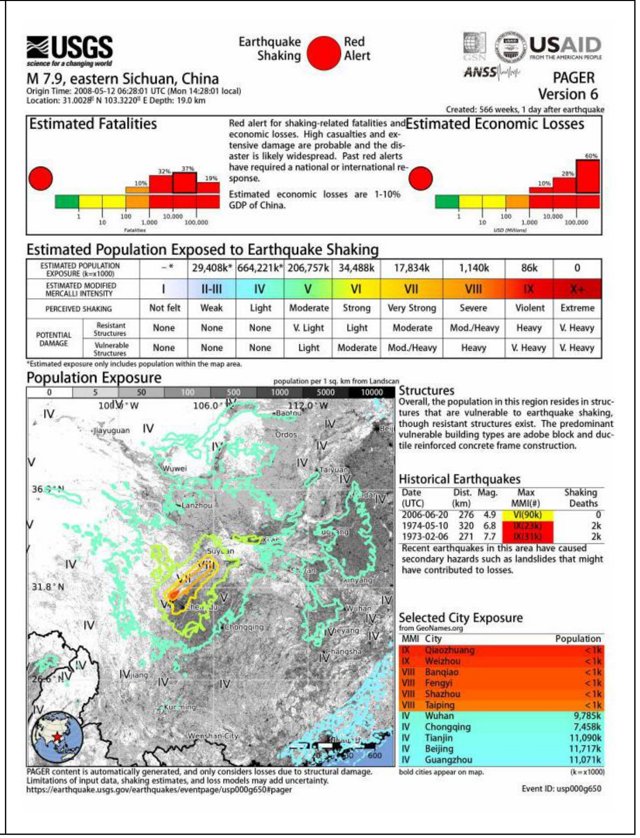


Figure S8. Fatalities and economic losses estimated by PAGER for the three source models.